End-to-End System Test Summary

1. Flight Search

I have implemented an end-to-end system test for the flight search functionality of the application. The test focuses on searching for a flight from Toronto (YYZ) to Vancouver (YVR) on August 6, 2024, verifying that all elements work correctly.

- Features tested:
    1. *Flight Search Form Inputs:* The test verifies that the "From" and "To" dropdown menus are working properly, allowing the user to select Toronto (YYZ) as the origin and Vancouver (YVR) as the destination.

    2. *Date Selection:* The date picker functionality is also tested. The script opens the date picker, selects the correct year (2024), month (August), and day (6th). This step ensures that the date picker accurately reflects user input and that the selections are correctly captured.

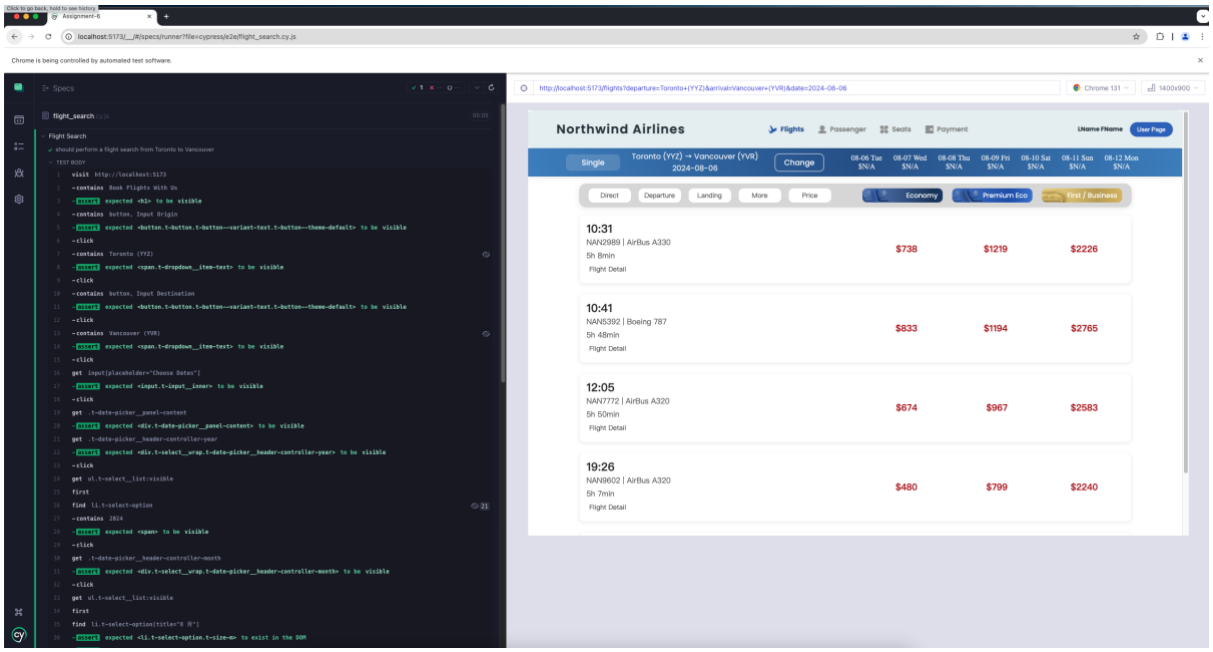
    3. *Flight Search Results Page:* The script verifies that clicking the "Search" button loads the correct results page, and checks the URL parameters to ensure they match the selected origin, destination, and date. Additionally, it validates that flight details (such as flight numbers, ticket prices, and class types) are displayed on the results page.

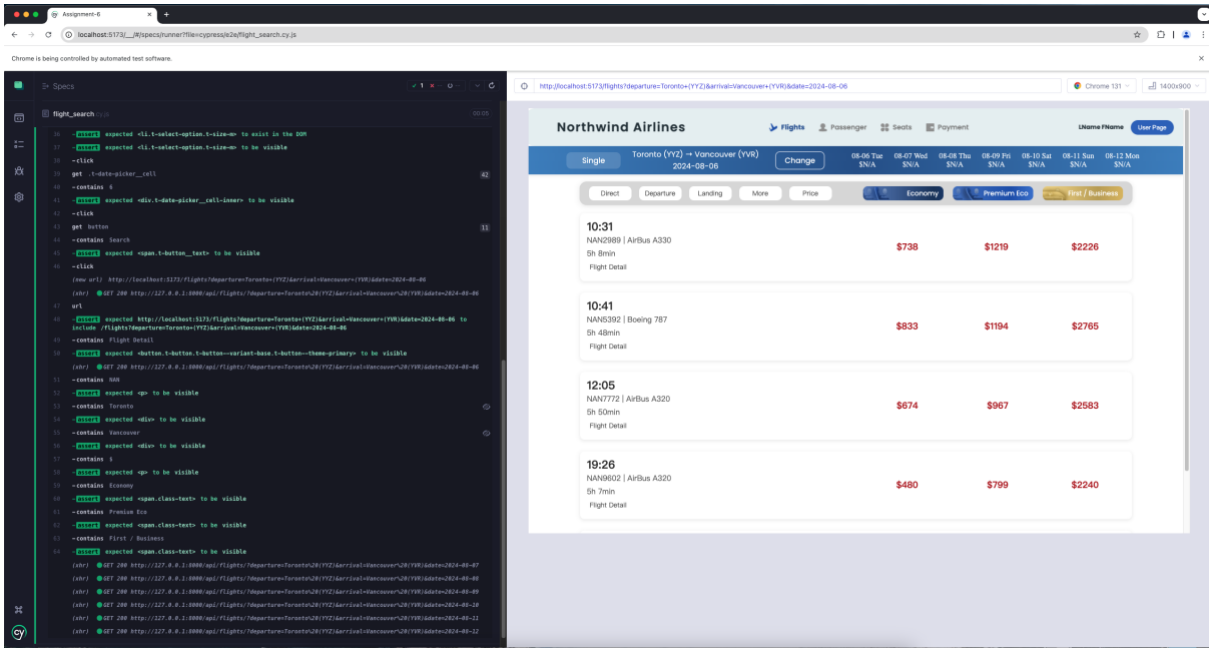
- Test Script Summary: The script was written in JavaScript using Cypress
    1. *Visiting the Main Page:* The script begins by navigating to the main page of the flight booking system and ensuring that the title "Book Flights With Us" is visible.
    2. *Selecting the Origin and Destination:* The script selects Toronto (YYZ) as the origin and Vancouver (YVR) as the destination using dropdown menus. This part of the test ensures that the selection process for flights works correctly.
    3. *Selecting the Date:* The script interacts with the date picker to select August 6, 2024, as the departure date. It ensures that the date picker UI is visible, functional, and the correct selections can be made.
    4. *Performing the Flight Search:* The script simulates clicking the "Search" button and verifies that the URL changes accordingly. It also ensures that the flight results are displayed with accurate details, such as flight numbers, prices, and available classes.

- Running the Test Script: To run the Cypress test script, follow these steps:

1. Before running this project, make sure you have the following installed on your system:
    - Node.js (v16.x or higher recommended)
    - npm (comes with Node.js)
2. Clone the repository:
    - git clone https://github.com/mentlesoul/CISC327-Group-22-TA.git
    - cd CISC327-Group-22-TA\Assignment-6
3. Installation dependencies: At the root directory, thus Assignment-6 folder, run bash/shell command below:
    - npm install
4. Local development launch:
    - npm run dev
5. Visit http://localhost:5173/ to view the page.
6. To run the Django backend service
    - cd backend
    - python manage.py runserver
7. Ensure that all dependencies for Cypress are installed in the project directory.
8. Open a terminal in the project directory and use one of the following commands to run the test script:
    - npx cypress open: Opens the Cypress Test Runner UI, where you can manually select the test to run.
    - npx cypress run: Runs the test script directly in the terminal, without the UI.

Result of flight_search.cy.js:





```
(Run Starting)

  Cypress:        13.16.0
  Browser:        Electron 118 (headless)
  Node Version:   v20.18.0 (/usr/local/bin/node)
  Specs:          1 found (flight_search.cy.js)
  Searched:       cypress/e2e/**/*.cy.{js,jsx,ts,tsx}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  Running:  flight_search.cy.js                                             (1 of 1)


  Flight Search
    ✓ should perform a flight search from Toronto to Vancouver (4190ms)


  1 passing (4s)


  (Results)

  ┌────────────────────────────────────────────────────────────────────────────┐
  │ Tests:        1                                                            │
  │ Passing:      1                                                            │
  │ Failing:      0                                                            │
  │ Pending:      0                                                            │
  │ Skipped:      0                                                            │
  │ Screenshots:  0                                                            │
  │ Video:        false                                                        │
  │ Duration:     4 seconds                                                    │
  │ Spec Ran:     flight_search.cy.js                                          │
  └────────────────────────────────────────────────────────────────────────────┘


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  (Run Finished)


       Spec                                              Tests  Passing  Failing  Pending  Skipped
  ┌────────────────────────────────────────────────────────────────────────────────────────────────┐
  │ ✓  flight_search.cy.js                      00:04        1        1        -        -        - │
  └────────────────────────────────────────────────────────────────────────────────────────────────┘
    ✓  All specs passed!                        00:04        1        1        -        -        -

○ (base) lamchinghei@ChingheideMacBook-Pro Assignment-6 %
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